



How Well Do Carbon Taxes Match Their Promise? Evidence from 2023

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Key Findings

- Carbon taxes are a popular solution to climate change because they are an economically efficient way to price in negative externalities of greenhouse gas emissions.
- In practice, carbon taxes have been partial measures, either exempting certain sectors of the economy or offering substantially reduced rates for certain activities.
- One way to judge the design and administration of carbon taxes is by measuring the ratio of actual revenue collected divided by potential revenue, known as a c-efficiency ratio. The resulting ratio illustrates how well the tax captures the theoretical tax base.
- Using emission, tax rate, and revenue data from 2023, we estimated the c-efficiency ratio of carbon taxes around the world.
- Luxembourg, Japan, and the Canadian jurisdictions of Northwest Territories and British Columbia performed the best, with c-efficiency ratios well above 0.6.
- The average c-efficiency ratio of the jurisdictions studied is 0.29, and the weighted average c-efficiency ratio is 0.26.
- Some countries employ multiple carbon pricing systems, often for different sectors of the economy, which explains some carbon taxes' lower c-efficiency ratios.
- To account for this, we also used an adjusted calculation that subtracted emissions excluded from carbon taxes but subject to other carbon pricing systems (most commonly the European Union's Emissions Trading System).
- After implementing this adjustment, c-efficiency ratios rise to an unweighted average of 0.36 and a weighted average of 0.36.

- While most jurisdictions fall short of the ideal, high c-efficiency ratios for some jurisdictions that rely heavily on carbon taxes show it is a viable policy tool for capturing its desired tax base.

Introduction

The carbon tax is often considered the ideal climate policy. However, existing carbon taxes do not match the ideal theoretical policy.[1] More than two dozen carbon taxes exist in the world today, and they vary widely in terms of design. Some cover only a limited share of emissions or feature reduced rates for different sectors.

The first edition of this report created a universal statistic to show how well existing carbon taxes compared to an idealized theoretical tax on greenhouse gas emissions.

The second edition of the report provides two versions of the same statistic: one that looks at carbon taxes alone, and another that considers the presence of other carbon prices in the given jurisdiction.

Methodology and Issues

C-Efficiency Ratio: The Premise

This paper applies the concept of c-efficiency ratios, commonly used to evaluate value-added taxes (VATs), to carbon taxes. The ratio compares the extent to which a tax raises revenue equal to the total theoretical consumption tax base multiplied by the tax rate. If a tax was perfectly designed and enforced, the efficiency ratio should equal 1.[2] In practice, most taxes have enforcement challenges that create compliance gaps and deliberate design choices, such as exemptions, that create policy gaps. Together, the compliance and policy gaps reduce the c-efficiency ratio below 1.[3] While some divergence from a theoretical ideal is to be expected—no tax has a perfect compliance rate and certain aspects of total consumption are particularly difficult to measure—policymakers have significant control over policy gaps.

Applied to value-added taxes in 2018, the unweighted average efficiency ratio in the OECD was 0.56, meaning slightly more than half of potential VAT revenue was collected.[4] Across individual countries, efficiency ratios ranged from 0.35 at the low end to 0.75 at the high end.[5] While a revenue ratio of 1 is an unrealistic expectation, an average ratio of 0.56 suggests that VAT bases have significant room for improvement.

As carbon taxes are supposed to be broad taxes on all greenhouse gas emissions, the revenue generated should be emissions times the carbon tax rate. Calculating the efficiency ratio for a carbon tax, by dividing actual revenue by potential revenue, allows us to determine how well carbon taxes capture the potential greenhouse gas emissions. This paper considers all carbon taxes active for the entirety of 2023 according to the World Bank's Carbon Pricing Dashboard with limited exceptions.[6]

What Drives the Ratio?

The carbon tax efficiency ratio has two components that lower it below 1: a compliance gap and a policy gap. The compliance gap reflects carbon tax revenue legally owed but not remitted to the government levying the tax. Meanwhile, the policy gap reflects revenue not collected because some emissions are either exempted from tax or taxed at a reduced rate under the law.

Sector and Industry Exemptions or Reduced Rates

Carbon emissions are classified according to the sector from which emissions arise: power generation, industry, transportation, residential and commercial buildings, and agricultural and land use changes.[7]

Many existing carbon taxes exempt emissions from certain sectors or industries.[8] A few reasons motivate the exemptions or reduced rates: some sectors may be subject to a different emissions reduction policy, governments may fear that a carbon tax would make a domestic sector or industry uncompetitive, or political constraints might make only a partial carbon tax on some sectors viable.

The most common intersecting policy with carbon taxes in this report is the European Union's Emissions Trading System, also known as the EU ETS. The ETS is a cap-and-trade system, where companies must pay for a fixed number of allowances for pollution (although some industries receive free allowances to preserve competitiveness). As of 2023, the EU ETS covered emissions from the power sector, certain emissions-intensive industrial activities, and intra-EU aviation, cumulatively covering roughly 40 percent of EU greenhouse gas emissions.^[9] Many countries have designed their domestic carbon taxes to complement the EU ETS, excluding emissions that are subject to the EU-level policy. EU countries also often impose significant excise taxes on various energy products.^[10]

De Minimis Exemptions

At some margin, the cost of administering and complying with a tax becomes higher than the revenue collected. As a result, many taxes (not just carbon taxes) have de minimis exemptions designed to prevent certain small entities from incurring a large compliance burden just to remit a trivial sum to the government treasury.

We should expect some amount of emissions to be excluded from a carbon tax base for de minimis exemptions. However, according to most analyses, most carbon emissions are capturable in a carbon tax. When administered upstream, estimates from Gilbert Metcalf and Jeffrey Weisbach found that around 80 percent of carbon emissions in the United States could be taxed, while covering only 3,000 taxpayers, and that up to 90 percent of carbon emissions could be taxed with modest additional cost.^[11]

A carbon tax is an excise tax, and most excise taxes have low administrative costs relative to the revenue collected, as relatively few taxpayers pay it directly.^[12] So, while reasonable de minimis exemptions would keep an ideal carbon tax efficiency ratio shy of 1, they should not be a primary driver of a low c-efficiency ratio.

Noncompliance

A carbon tax would include some level of noncompliance. While few studies exist on the range of noncompliance rates for carbon taxes, other studies provide enough context to determine a reasonable range. In aggregate, the United States collects roughly 85 percent of taxes legally owed voluntarily, and 87 percent of taxes owed once considering late payments

and enforcement collections.[13] This translates to a gross tax gap of 15 percent and a net tax gap of 13 percent in the United States. When looking at excise taxes in the United States, the tax gap is much lower on a rate basis. The US federal government collected an average of \$95 billion in excise taxes between 2014 and 2016, and the average excise tax gap was under \$0.5 billion, translating to a tax gap rate below 1 percent.[14]

Meanwhile, in Europe, the compliance gap for value-added taxes is roughly 9.1 percent of expected VAT revenue.[15] For our purposes, determining the carbon tax compliance gap is unnecessary.

Challenges to Finding the Right Carbon Tax Base

While it may seem straightforward to determine the proper carbon tax base, a series of challenges complicate the effort, including how to account for intersecting policies, the selection of greenhouse gases, and the question of border adjustments.

Intersecting Policies

Some countries have multiple carbon pricing policies. For example, many European countries have national-level carbon taxes, while also being subject to the EU's Emissions Trading System. As a result, many countries exempt sectors already subject to the other policies. In other cases, countries have sector-specific excise taxes on, for example, motor fuel or electricity use. Such excise taxes are not based on carbon emissions but may be partially aimed at targeting social costs related to carbon emissions. Furthermore, countries may choose more direct government involvement through either regulation or subsidy as a solution in some sectors.

Situations like these create a problem for measuring a carbon tax efficiency ratio. Consider a country that has a carbon tax of \$40 per ton and participates in the EU ETS. The ETS covers emissions from electricity generation, among other sectors. In 2021, the ETS's average price was roughly \$65 per ton. If the country includes electricity generation in the carbon tax base, the resulting carbon emissions would face a combined tax of \$105 per ton. That would be an excessive tax burden relative to emissions that don't also face the ETS, so it would make sense to exclude the electricity generation sector from the carbon tax.

The initial version of this paper set the issue of multiple carbon prices active in one jurisdiction aside. Instead, it solely considered carbon taxes in isolation. Having multiple carbon pricing policies is not ideal: part of the appeal of the carbon tax is that it is universal, and relying on multiple policies which (usually) have different carbon prices runs contrary to that premise. The qualitative section of the paper provided additional context discussing whether a carbon tax's low score resulted from it acting as a complementary instrument with another carbon price like the EU ETS.

In this updated edition, we have expanded the analysis to include two c-efficiency scores. The unadjusted score is based on the carbon tax alone, calculating potential revenue by multiplying total jurisdiction emissions by the highest carbon tax rate in the jurisdiction. The adjusted score considers other active carbon pricing mechanisms. Instead of basing the c-efficiency score on the carbon tax rate times total jurisdictional emissions, it is calculated based on jurisdictional emissions minus jurisdictional emissions subject to another carbon price.

The adjusted c-efficiency calculation is not perfect: in many cases, the other carbon price is not set at the same level as the carbon tax, which undermines the promise of a carbon tax's neutrality across different types of emissions. The adjustment calculation also does not address the relatively marginal cases when the carbon tax and the other carbon price are mostly complementary, but some small share of emissions is subject to both carbon pricing systems. Additionally, it does not consider excise taxes on goods like motor fuel that are sometimes set at a rate in excess of road maintenance costs in order to serve a climate goal, as those taxes do not meet a strict definition of carbon tax. It would be ideal if all emissions in a region were subject to one uniform carbon tax. However, excluding a sector from a carbon tax because it is already subject to a different carbon pricing system is more justifiable than excluding a sector without any existing carbon price.

“Carbon” Emissions versus Greenhouse Gases

Carbon dioxide emissions (or carbon emissions, in shorthand) are usually the focus of climate policy, with carbon emissions often used interchangeably with greenhouse gas emissions. However, carbon dioxide is not the only greenhouse gas of note. Methane and nitrous oxide also contribute to climate change through the same channels that carbon

dioxide does.[16] While the volumes of methane and nitrous oxide are quite low, they are more powerful than CO₂ in terms of their impact on the climate on a per-volume basis, so greenhouse gas emissions are often measured in CO₂ equivalents. In 2021, CO₂ was responsible for nearly 80 percent of gross US greenhouse gas emissions as measured in CO₂ equivalents, while methane was responsible for 11.5 percent of emissions, nitrous oxide for 6.2 percent, and a handful of fluorinated gases for the remaining 3 percent.[17]

Ideally, a “carbon” tax would cover all greenhouse gas emissions, with rates scaling to the impact of the gas on climate change. In practice, some gases pose administrative challenges.[18] Large shares of nitrous oxide and methane emissions in the United States come from agricultural activities and are difficult to measure.[19] In some cases, a country’s emissions profile may skew towards non-energy, non-CO₂ emissions. As a result, the country may exclude a larger share of emissions (and thus receive a worse c-efficiency ratio) thanks to its more administratively challenging emissions profile.

Nonetheless, we include all greenhouse gas emissions, rather than just CO₂, as the idealized tax base. We will note unusual emissions profiles in the country-level discussion.

Production versus Consumption and Foreign Markets

An ideal carbon tax would be based on consumption, rather than production. A carbon tax based on the production of emissions creates a disadvantage for domestic industry. In the domestic market, a domestic firm must pay taxes on their emissions, while imported goods from foreign manufacturers would go untaxed. Conversely, in foreign markets, the domestic exporter still owes the carbon tax, and the foreign manufacturer is still untaxed.[20]

The imbalance can be solved by moving to a consumption base. Under a consumption-based carbon tax, the carbon content of imports is taxed, while the carbon content of exports is exempted. As a result, the domestic market is neutral between domestic and foreign producers (because both would owe the carbon tax), as is the foreign market (because neither would owe the carbon tax).[21]

Introducing a border adjustment poses several challenges, most acutely measuring the carbon content of imported goods.^[22] While it is possible to overcome the challenges, most carbon taxes follow a production base rather than a consumption base. As a result, we will use production emissions, even though ideally a carbon tax would be based on emissions from consumption.

Data Sources, Methodology, and Challenges

We obtain data on carbon tax rates and actual revenue collected from the World Bank's Carbon Pricing Dashboard on carbon taxes around the world, including both taxes applied nationally and taxes applied subnationally.^[23] Greenhouse gas emissions data come from EDGAR, except in the case of the Canadian provinces, where we used values from Canadian provinces' greenhouse gas emissions inventory reports.^[24] We calculate the c-efficiency ratio by dividing actual carbon tax revenue collected by the highest tax rate on CO₂ emissions multiplied by the jurisdiction's emissions. One notable methodological change from the last edition is going from using the primary carbon tax rate to using the highest carbon tax rate. In most cases, these values are the same: carbon taxes typically have a standard rate that is also the highest, but there are a few jurisdictions with a special extra rate for certain types of emissions. We adjust nominal price and revenue estimates with exchange rates as of March 31, 2024, per the World Bank's database.^[25]

For each country, we used revenue and rate data from 2023.

Another possible source of mismeasurement is midyear changes in carbon tax rates. The c-efficiency ratio calculations use rates as of April 1, 2023.^[26] The underlying assumption is that the period that rate is in place matches up with the time period the actual 2023 revenue was collected. Midyear changes in the carbon tax rate could lead to an overestimate or underestimate of the tax's efficiency. If a carbon tax is raised midyear, then the estimated efficiency ratio will be an underestimate; if the tax is reduced midyear, the estimated efficiency ratio will be an overestimate.^[27] We have noted the jurisdictions in which such an issue could be present.

2023 Carbon Tax C-Efficiency Ratios

The c-efficiency ratios range from a high of 0.78 in Luxembourg to a low of Poland's at 0.00. The average of all countries studied was 0.29. The weighted average c-efficiency ratio (calculated by adding up all revenue collected and dividing by the sum of each country's potential revenue, so effectively weighted by country emissions) was 0.26.

The picture improves when adjusting for other carbon pricing systems active in a jurisdiction. The unweighted average rises to 0.36 after subtracting out the bases of complementary or mostly complementary carbon price systems. The weighted average, meanwhile, also rises to 0.36. Luxembourg has the highest adjusted c-efficiency ratio: 0.88.

In the table on each country's page, we present revenue and rate data for each country in terms of US dollars, adjusted using exchange rates as of March 31, 2024. However, in the text, we often discuss past tax rates or planned future tax rates in terms of local currency. This is to avoid misleading situations where, thanks to fluctuating exchange rates, a carbon tax in a country may have fallen over time when converted to dollars, but in local currency terms rose.

Table 1. 2023 Carbon Tax C-Efficiency Ratios

Country	Unadjusted Rank	Unadjusted C-Efficiency Ratio	Adjusted Rank	Adjusted C-Efficiency Ratio	Other Carbon Price Active in Jurisdiction?
Luxembourg	1	0.78	1	0.88	Yes, Complementary
Japan	2	0.74	2	0.74	Yes, Not Complementary
British Columbia	3	0.67	4	0.67	No
Northwest Territories	4	0.66	5	0.66	No
Singapore	5	0.58	8	0.58	No
Ukraine	6	0.48	12	0.48	No

Country	Unadjusted Rank	Unadjusted C-Efficiency Ratio	Adjusted Rank	Adjusted C-Efficiency Ratio	Other Carbon Price Active in Jurisdiction?
France	7	0.45	10	0.55	Yes, Complementary
Denmark	8	0.43	9	0.55	Yes, Complementary
Germany	9	0.41	3	0.72	Yes, Complementary
Finland	10	0.39	7	0.60	Yes, Complementary
Austria	11	0.36	11	0.53	Yes, Complementary
Iceland	12	0.35	6	0.61	Yes, Complementary
Ireland	13	0.33	15	0.42	Yes, Complementary
Portugal	14	0.32	16	0.42	Yes, Complementary
Sweden	15	0.31	13	0.47	Yes, Complementary
Norway	16	0.28	14	0.45	Yes, Complementary
Liechtenstein	17	0.23	19	0.23	Yes, Complementary, No Data
Chile	18	0.23	20	0.23	No
Switzerland	19	0.21	18	0.23	Yes, Complementary
Canada	20	0.16	17	0.34	Yes, Complementary
Mexico	21	0.15	21	0.15	Yes, Mixed, No Data
United Kingdom	22	0.12	22	0.12	Yes, Not Complementary
Colombia	23	0.11	23	0.11	No

Country	Unadjusted Rank	Unadjusted C-Efficiency Ratio	Adjusted Rank	Adjusted C-Efficiency Ratio	Other Carbon Price Active in Jurisdiction?
Argentina	24	0.07	25	0.07	No
Estonia	25	0.05	24	0.09	Yes, Complementary
Hungary	26	0.05	26	0.06	Yes, Supplementary
Latvia	27	0.05	27	0.05	Yes, Complementary
Uruguay	28	0.04	28	0.04	No
South Africa	29	0.03	30	0.03	No
Spain	30	0.02	29	0.03	Yes, Complementary
Poland	31	0.00	31	0.00	Yes, Complementary
Weighted Average		0.26		0.36	
Unweighted Average		0.29		0.36	

Note: Albanian, Dutch, and Mexican subnational carbon taxes not included due to data sufficiency reasons. Provincial Canadian carbon taxes in Newfoundland and Labrador, Prince Edward Island, and New Brunswick excluded as they were supplanted by Canada's federal fuel charge during 2023. Slovenia excluded because the carbon tax was reintroduced in mid-2023. Meanwhile, the national Canadian carbon tax along with the carbon taxes in Northwest Territories and British Columbia are still included as they were fully active in 2023 even though they have since been either repealed or significantly curtailed.

Source: World Bank, "Carbon Pricing Dashboard," updated April 1, 2024; EDGAR, Joint Research Centre and International Energy Agency, "GHG Emissions of All World Countries: 2024 Report"; EU Emissions Trading System (ETS) Data Viewer, accessed December 2024, <https://www.eea.europa.eu/en/analysis/maps-and-charts/emissions-trading-viewer-1-dashboards>; Government of Canada, "Canada's Official Greenhouse Gas Inventory," accessed December 2024, <https://data-donnees.az.ec.gc.ca/data/substances/monitor/canada-s-official-greenhouse-gas-inventory/A-IPCC-Sector?lang=en>; German nEHS Administration.

Conclusion

The unweighted and weighted average c-efficiency ratios of 0.29 and 0.26 suggests that, on average, carbon taxes capture a limited share of their theoretical tax base. Even when adjusting for the presence of complementary carbon pricing systems, the average c-

efficiency ratio of 0.36 remains well short of even the average VAT c-efficiency ratio.

But both the unadjusted and adjusted averages are brought down by edge cases like Spain, South Africa, and Poland. Those three countries all have substantial emissions, and their carbon taxes either have very limited scope or, in the case of South Africa, are still in a transitional or rollout period of sorts.

The data also show proof of concept for carbon taxes that provide broad coverage of emissions at consistent rates and are administered effectively. Luxembourg tops both the unadjusted and adjusted rankings with c-efficiency ratios higher than even very good VAT systems. Jurisdictions like Japan and the Canadian regions of Northwest Territories and British Columbia show the viability of a carbon tax-only approach.

The number of carbon taxes with c-efficiency ratios above the average VAT c-efficiency ratio grows substantially after accounting for carbon taxes that serve as complements for other carbon pricing systems. Before adjusting for other carbon pricing systems, only five jurisdictions have c-efficiency ratios that exceed the average OECD VAT c-efficiency ratio of 0.56. However, when considering adjusted c-efficiency ratios, eight jurisdictions exceed 0.56, with France and Denmark barely missing that threshold with adjusted c-efficiency ratios of 0.55.

The results underscore some similarities between VATs and carbon taxes internationally. Both types of taxes have strong theoretical cases. Most international examples of both fall short of the optimal theoretical case, but nonetheless contribute to their respective goals. While VATs may not perfectly match the theoretical arguments that they tax all consumption equally, in practice, existing VATs are still less economically harmful means of generating revenue than income taxes.^[28] Similarly, while carbon taxes might not perfectly match the theoretical ideal case, they can incentivize reductions in emissions and raise revenue used to reduce other taxes.^[29]

At the same time, a low c-efficiency score relative to other comparable countries suggests policymakers have significant room for improving the design of their respective carbon taxes to accurately and equally price emissions. And in an ideal world, countries would move

towards a single carbon price, instead of relying on separate policies for different economic sectors.

Carbon Taxes by Country or Region

Argentina

Argentina's Carbon Tax System

Rate (2023)	\$35.07 per ton
Total GHG Emissions (2023)	72.92 megatons of CO2-equivalent
Actual Revenue (2023)	\$0.91 billion
Potential Revenue	\$2.56 billion
Unadjusted C-Efficiency Ratio	0.36 (11th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	24.41 megatons of CO2-equivalent
Adjusted Potential Revenue	\$1.70 billion
Adjusted C-Efficiency Ratio	0.53 (11th of 31)

Source: Author's calculations; World Bank Carbon Pricing Dashboard; EDGAR, "GHG Emissions of All Countries, 2024," September 2024.

Argentina introduced a carbon tax in 2018. The tax replaced existing excise taxes on motor fuel, including gas and diesel taxes, and a separate tax on imported fuel. Meanwhile, fuels such as coal and petroleum coke were subject to an initially lower carbon tax rate per ton that was intended to increase incrementally over time.^[30] Updates intended to bring the rates on different fuels closer to parity were scheduled to begin in the second half of 2021, but were postponed several times, eventually taking effect in February 2024.^[31] Argentina is also considering adopting an ETS; it was initially included in a government omnibus proposal in December 2023, but was withdrawn from a later version of the same package.^[32]

A narrow scope is the major driver of the Argentine carbon tax’s low score. The most important gap in the Argentine tax base is non-CO2 greenhouse gas emissions like methane and nitrous oxide. Mostly coming from the agricultural sector, non-CO2 GHGs make up almost half of Argentina’s total emissions as of 2023.^[33] Additionally, the carbon tax exempts CO2 emissions from natural gas, which provides almost half of the country’s energy generation.^[34] Beyond the significant exemptions, the tax also featured lower rates for mineral coal and petroleum coke in 2023.

Austria

Austria's Carbon Tax System

Rate (2023)	\$35.07 per ton
Total GHG Emissions (2023)	72.92 megatons of CO2-equivalent
Actual Revenue (2023)	\$0.91 billion
Potential Revenue	\$2.56 billion
Unadjusted C-Efficiency Ratio	0.36 (11th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	24.41 megatons of CO2-equivalent
Adjusted Potential Revenue	\$1.70 billion
Adjusted C-Efficiency Ratio	0.53 (11th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024.

Austria had a carbon tax as of 2023, but the policy is typically described as an ETS. First introduced in 2022, the Austrian government imposed a fixed carbon price (in practice a carbon tax) as a transitional policy as it prepared a more complex ETS. In 2026, the policy instrument will be converted to an ETS. The Austrian carbon price is also complementary with the EU’s ETS, and the Austrian price is scheduled to be supplanted by EU ETS 2, a new EU-level policy designed to cover emissions not subject to the original EU ETS, whenever EU ETS 2 takes effect.^[35]

Even though it is a complementary instrument, Austria’s carbon tax performs well under the c-efficiency ratio regardless. It covers emissions from non-ETS sectors, such as buildings and road transportation, as well as any emissions from power generation not subject to the ETS.[36] Transportation is the largest source of emissions in Austria, so covering transportation comprehensively is the main source of the Austrian policy’s strength. The policy is focused on fuel and energy use, so it is missing some emissions from agriculture.

The Austrian carbon pricing system looks even stronger after adjusting for the EU ETS’s coverage in the country.

British Columbia

British Columbia's Carbon Tax System

Rate (2023)	\$48.03 per ton
Total GHG Emissions (2023)	60.48 megatons of CO2-equivalent
Actual Revenue (2023)	\$1.96 billion
Potential Revenue	\$2.90 billion
Unadjusted C-Efficiency Ratio	0.67 (3rd of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.67 (4th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; Environment and Climate Change Canada Data Catalogue, “Environmental Provincial and Territorial Greenhouse Gas Emission Tables by IPCC Sector, 1990–2022.”

Established in 2008, British Columbia’s carbon tax is often considered one of the best examples of the policy.[37] The tax was introduced at a low rate of CAD 10 per ton in 2008, before steadily increasing by CAD 5 per ton until reaching CAD 30 per ton by 2012.[38] The tax is applied to almost all CO2 emissions, and the tax revenue has been used to reduce marginal corporate and individual tax rates and provide transfer payments to lower-income

households.^[39] In 2023, the tax rate was CAD 65. Starting in 2024 (after our 2023 timeframe), industrial operations were subject to a cap-and-trade scheme known as the British Columbia Output-Based Pricing System (OBPS) instead of the carbon tax.^[40]

The high c-efficiency ratio of British Columbia’s carbon tax means the tax lived up to its reputation for being well-designed and broad. The main drivers of its high rating are the broad base and uniform tax rate. The tax’s exemptions are quite narrow, notably including a rudimentary border adjustment exempting the emissions content of exported fuels.^[41] The tax includes other greenhouse gas emissions like methane and nitrous oxide, as opposed to just CO2.^[42] The British Columbian carbon tax was one of the best examples of a well-designed carbon tax. However, the tax was dropped to zero in April 2025.^[43]

Canada

Canada's Carbon Tax System

Rate (2023)	\$48.03 per ton
Total GHG Emissions (2023)	747.68 megatons of CO2-equivalent
Actual Revenue (2023)	\$5.72 billion
Potential Revenue	\$35.91 billion
Unadjusted C-Efficiency Ratio	0.16 (20th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	401.06 megatons of CO2-equivalent
Adjusted Potential Revenue	\$16.65 billion
Adjusted C-Efficiency Ratio	0.34 (17th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; Environment and Climate Change Canada Data Catalogue, “Environmental Provincial and Territorial Greenhouse Gas Emission Tables by IPCC Sector, 1990–2023.”

Canada’s federal carbon tax (also known as the federal fuel charge) was introduced in 2019, and the minimal rate was set at CAD 20 and increased by CAD 10 until reaching CAD 50 per ton in 2022.^[44] It was increased by another CAD 15 to CAD 65 in 2023 and CAD 80 in 2024.

Canada repealed the federal tax in 2025.^[45] The tax applied in cases where a province’s carbon pricing policy did not meet the stringency requirements of the federal government, serving as a national level backstop. The federal fuel charge was not the only national-level carbon pricing backstop, as Canada still has a national-level cap-and-trade system for major emitters.

The national-level Canadian carbon tax performs poorly under the c-efficiency ratio because emissions in several provinces were excluded from the tax base, as they had their own provincial-level carbon taxes, as well the separate cap-and-trade systems for industry. In 2023, Quebec, Northwest Territories, and British Columbia were fully exempt from federal carbon pricing initiatives. The federal fuel charge applied in Alberta, Saskatchewan, Ontario, New Brunswick, Nova Scotia, and Newfoundland and Labrador, alongside provincial cap-and-trade systems. The federal fuel charge applies alongside the federal cap-and-trade system in Manitoba, Prince Edward Island, Nunavut, and Yukon.^[46] Over half of Canadian emissions are covered by other carbon pricing systems.

Even after taking those design choices into account, the c-efficiency ratio is still low. Part of this difference may be a timing issue: the federal fuel charge only started applying in certain regions—namely New Brunswick, Nova Scotia, and Newfoundland and Labrador—in the middle of 2023, so the tax only applied there for half of the year. Non-CO2 GHG emissions from agriculture were also excluded.

Going forward, these design considerations are moot: Canada repealed its federal carbon tax in April 2025.^[47]

Chile

Chile's Carbon Tax System

Rate (2023)	\$5.00 per ton
Total GHG Emissions (2023)	121.46 megatons of CO2-equivalent
Actual Revenue (2023)	\$139.00 million

Rate (2023)	\$5.00 per ton
Potential Revenue	\$607.32 million
Unadjusted C-Efficiency Ratio	0.23 (18th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.23 (20th of 31)

Source: Author's calculations; World Bank Carbon Pricing Dashboard; EDGAR, "GHG Emissions of All Countries, 2024," September 2024.

Chile's carbon tax system was passed in 2014 as part of a broader tax reform and went into effect in 2017.^[48] The system includes an annual tax on emissions from fixed sources, targeting CO₂ emissions from boilers and turbines with a thermal power of 50MW or more, ultimately covering 93 facilities as of 2018.^[49] The initial aim of the carbon tax was to help Chile achieve its climate mitigation goal of reducing carbon emissions by 20 percent by 2020 compared to 2007 levels. Another tax reform passed in 2020 included reforms to the tax base scheduled to take effect at the beginning of 2023.^[50] The Chilean carbon tax has two components: a conventional carbon tax and a fee on local particulate pollution.^[51]

Chile's carbon tax c-efficiency ratio is roughly middle-of-the-pack. The two largest emissions-producing sectors in Chile are power (largely covered by the tax) and transportation (largely exempt from it).^[52] The tax also does not cover non-CO₂ GHG emissions, which combined to account for around 30 percent of emissions in 2023.^[53] Reforms taking effect at the beginning of 2023, which expanded the tax base by lowering the de minimis threshold to facilities with over 25,000 tons of CO₂ and those that release over 100 tons of particulate matter, should improve Chile's carbon tax efficiency rating going forward.^[54]

Colombia

Colombia's Carbon Tax System

Rate (2023)	\$5.06 per ton
Total GHG Emissions (2023)	223.97 megatons of CO ₂ -equivalent
Actual Revenue (2023)	\$0.12 billion
Potential Revenue	\$1.13 billion
Unadjusted C-Efficiency Ratio	0.11 (23rd of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.11 (23rd of 31)

Source: Author's calculations; World Bank Carbon Pricing Dashboard; EDGAR, "GHG Emissions of All Countries, 2024," September 2024.

Enacted in 2017, the Colombian carbon tax operates as a levy on the carbon content within liquid and gaseous fossil fuels, encompassing fuels used for transportation, stationary power generation, or heating.^[55] The tax only covers natural gas when the gas is used in refineries or the petrochemical sector. The tax initially excluded coal, but a tax reform signed in late 2022 expanded the carbon tax's base to coal, slowly phasing in a carbon tax on coal from 25 percent of the carbon tax rate in 2025 to 100 percent by 2027.^[56] The reform also included an annual inflation adjustment to the carbon tax rate.^[57]

Colombia's carbon tax has a relatively low c-efficiency ratio. The most significant hole in the tax base is non-CO₂ GHG emissions. In 2023, over half of Colombia's GHG emissions were non-CO₂ gases, such as methane and nitrous oxide.^[58] The exclusion for coal is also a poor policy choice, but coal does not play a particularly important role in Colombia's energy mix, so the exclusion has a small effect on the nation's c-efficiency ratio.^[59] The wide availability of carbon offsets that allow companies to legally avoid paying the carbon tax through investment in some net-carbon-negative project (typically some form of reforestation) may also play a role in reducing the tax's coverage.^[60]

Denmark

Denmark's Carbon Tax System

Rate (2023)	\$26.59 per ton
Total GHG Emissions (2023)	41.83 megatons of CO2-equivalent
Actual Revenue (2023)	\$0.48 billion
Potential Revenue	\$1.11 billion
Unadjusted C-Efficiency Ratio	0.43 (8th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	9.24 megatons of CO2-equivalent
Adjusted Potential Revenue	\$0.86 billion
Adjusted C-Efficiency Ratio	0.55 (9th of 31)

Source: Author's calculations; World Bank Carbon Pricing Dashboard; EDGAR, "GHG Emissions of All Countries, 2024," September 2024; EU Emissions Trading System (ETS) Data Viewer.

Denmark introduced a carbon tax on energy from carbon-intensive sources in 1992 as part of a larger environmental package that also included energy taxes, a sulfur tax, and green subsidies.^[61] In 2023, the tax was levied at a rate of just over DKK 180 per ton; by 2030, Denmark plans to raise the rate to DKK 750 per ton for companies outside of the EU ETS, to DKK 375 per ton for companies in the EU ETS, and to DKK 125 for emissions from certain types of industrial processes.^[62]

The tax covers carbon emissions from gasoline and petroleum coke and captures mainly emissions from the buildings and transport sectors. The use of carbon tax reimbursements for emission reductions relative to historical fuel use under competitiveness considerations decreases Denmark's rank.^[63] While the tax does not currently cover agricultural methane (which accounts for 15 percent of the country's GHG emissions as of 2023), the Danish government has announced plans to tax livestock emissions starting in 2030.^[64]

Denmark’s carbon tax covers a large share of its emissions, which explains its relatively high rank. Most operators subject to the EU ETS are exempted, except for central heating.^[65] Subtracting out ETS emissions improves Denmark’s c-efficiency rating, although that improvement is somewhat deceptive as some of those emissions are still subject to the carbon tax.

Estonia

Estonia's Carbon Tax System

Rate (2023)	\$2.18 per ton
Total GHG Emissions (2023)	14.36 megatons of CO2-equivalent
Actual Revenue (2023)	\$1.7 million
Potential Revenue	\$31.2 million
Unadjusted C-Efficiency Ratio	0.05 (25th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	5.26 megatons of CO2-equivalent
Adjusted Potential Revenue	\$19.8 million
Adjusted C-Efficiency Ratio	0.09 (24th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer.

Estonia first introduced a small carbon tax in 2000, focused on thermal energy producers with capacity over 50 MW. A reform introduced in 2006 applied the tax to all thermal energy producers.^[66] The tax rate grew steadily from EUR 0.32 per ton to EUR 2 per ton, and increased to 25 EUR in July 2024.^[67] Estonia is also introducing a tax on motor vehicles partially based on carbon emissions.^[68] It is worth noting that Estonia has numerous other environmental charges, such as a high excise tax on motor fuel, which combine to raise significant revenue.^[69]

The Estonian carbon tax’s low c-efficiency ratio can be predominantly explained by both narrow scope and substantial targeted exemptions. For one, it does not include transportation, which accounts for a sixth of emissions in 2023. It also does not include non-CO2 emissions, which account for over 20 percent of emissions, and even in the power sector major producers can receive exemptions for retrofitting.^[70] While subtracting out emissions covered by the EU ETS does lead to a higher adjusted c-efficiency ratio, some facilities paid both the ETS and the Estonian carbon tax in 2023. As of 2024 though, the Estonian carbon tax is complementary with the EU ETS, so producers subject to the EU ETS will be exempt from the carbon tax.^[71]

Finland

Finland's Carbon Tax System

Rate (2023)	\$83.74 per ton
Total GHG Emissions (2023)	43.45 megatons of CO2-equivalent
Actual Revenue (2023)	\$1.42 billion
Potential Revenue	\$3.64 billion
Unadjusted C-Efficiency Ratio	0.39 (10th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	15.37 megatons of CO2-equivalent
Adjusted Potential Revenue	\$2.35 billion
Adjusted C-Efficiency Ratio	0.60 (7th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer.

Finland was the first country in the world to introduce a carbon tax, doing so in 1990. However, the tax was low initially.^[72] Finland’s carbon tax is part of the country’s overall excise tax on liquid fuels.^[73] Reforms in 1997, 2007, and 2011 significantly raised the tax while adjusting Finland’s excise tax on energy at the same time, and the carbon tax rate has been continually ratcheted up since then, amounting to EUR 77 in 2023.

Finland’s above-average c-efficiency ratio can be credited to its broad scope, which includes the taxation of two main sources: heating and transportation. Electricity and heat generation is the largest emissions sector in Finland, and while electricity is exempt from the carbon tax, heat generation is not. Emissions from combined heat and power plants are covered, but at a lower rate. Meanwhile, the transportation sector is the second-largest sector in terms of emissions, and it is fully taxed.^[74] However, the tax excludes non-CO2 emissions, which constitute over a quarter of Finland’s emissions as of 2023.^[75]

The ETS covers over a third of Finnish emissions, and the carbon tax is mostly complementary to the ETS. Subtracting out ETS emissions improves Finland’s c-efficiency ratio, but that adjustment is slightly overstated due to the ETS and carbon tax overlap in heat generation.^[76]

France

France's Carbon Tax System (2021)

Rate (2023)	\$48.50 per ton
Total GHG Emissions (2023)	385.52 megatons of CO2-equivalent
Actual Revenue (2023)	\$8.37 billion
Potential Revenue	\$18.70 billion
Unadjusted C-Efficiency Ratio	0.45 (7th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	70.60 megatons of CO2-equivalent
Adjusted Potential Revenue	\$15.27 billion
Adjusted C-Efficiency Ratio	0.55 (10th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer, German nEHS administration.

France introduced a carbon tax in 2014. The rate was set at EUR 7 per ton in 2014 and rose to EUR 44.6 per ton by 2018. That year, President Emmanuel Macron had planned to increase the tax further to around EUR 86 per ton by 2022, but political backlash as a part of the “Yellow Vest” movement led the French government to keep the carbon tax rate around EUR 44, which is where it remains.^[77] Despite the setback, the French carbon tax is still around the EU average.^[78]

The French carbon tax has a strong c-efficiency ratio thanks to its coverage of the transportation sector, which is responsible for over 30 percent of France’s GHG emissions.^[79] The tax only has limited exemptions and reduced rates in the transportation sector for commercial aviation, freight transport, and shipping. It also covers emissions from heating.^[80] The tax is limited to CO2, excluding methane, nitrous oxide, and HFCs. That has a substantial negative effect on the c-efficiency ratio, as methane and nitrous oxide emissions from agriculture make up around a sixth of the country’s overall GHGs.^[81]

France’s carbon tax is also designed as a complement to the EU ETS, which covers emissions in the power generation and industrial sectors.^[82] The ETS’s coverage is relatively small in France, as France relies heavily on nuclear energy rather than fossil fuels to generate electricity.^[83] Nonetheless, subtracting out emissions subject to the ETS produces a substantial improvement in the c-efficiency ratio.

Germany

Germany's Carbon Tax System

Rate (2023)	◆	\$32.37 per ton	◆
Total GHG Emissions (2023)		681.81 megatons of CO2-equivalent	
Actual Revenue (2023)		\$9.16 billion	
Potential Revenue		\$22.07 billion	
Unadjusted C-Efficiency Ratio		0.41 (9th of 31)	
GHG Emissions Subject to Other, Complementary Carbon Price		289.35 megatons of CO2-equivalent	

Rate (2023)	\$32.37 per ton
Adjusted Potential Revenue	\$12.70 billion
Adjusted C-Efficiency Ratio	0.72 (3rd of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer, German nEHS administration.

On paper, Germany does not have a carbon tax—its domestic carbon price is described as an emissions trading scheme. However, since it was first introduced, this policy (known as the nEHS) has functioned as a carbon tax, as the price has been fixed (adjusted each year, but not varying like a cap and trade system would).[84] The German national cap and trade system has been in place since 2021. It is intended as a complement to the EU ETS and covers emissions from sectors like transportation and residential and commercial buildings.

The national German carbon price performs well under the c-efficiency ratio even though it is only a complementary instrument. CO2 emissions from the transportation and building sectors alone account for almost 40 percent of all German greenhouse gas emissions.[85] The relatively high c-efficiency ratio also suggests effective enforcement and collection. CO2 emissions from the power sector are subject to the EU ETS and are therefore exempt from Germany’s national carbon price. Other than the power sector, the biggest hole in Germany’s national carbon price is non-CO2 emissions like methane, nitrous oxide, and F-gases. Combined, those gases accounted for over 14 percent of Germany’s total emissions in 2023. [86]

Subtracting out the emissions subject to the EU ETS substantially boosts the c-efficiency ratio of Germany’s national carbon price. The adjusted c-efficiency ratio of 0.72 reflects comprehensive coverage of greenhouse gas emissions with the exception of the aforementioned non-CO2 gases.

Hungary

Rate (2023)	\$38.69 per ton
Total GHG Emissions (2023)	60.93 megatons of CO ₂ -equivalent
Actual Revenue (2023)	\$0.12 billion
Potential Revenue	\$2.36 billion
Unadjusted C-Efficiency Ratio	0.05 (26th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	13.44 megatons of CO ₂ -equivalent
Adjusted Potential Revenue	\$1.84 billion
Adjusted C-Efficiency Ratio	0.06 (26th of 31)

Hungary introduced a carbon tax in January 2023. The Hungarian design is unusual; instead of being a complement to the EU ETS, it is a supplement to it. The tax falls on large facilities that produce an average of over 25,000 tons of CO₂ emissions during the preceding three years that also receive free EU ETS allowances in excess of 50 percent of their total emissions in the preceding year.^[87]

Given the tax's role as a narrow supplement to the EU ETS, Hungary's carbon tax has a low c-efficiency ratio. The EU ETS only covered 13.4 megatons of CO₂-equivalent emissions in 2023, making up less than a quarter of the nation's total.^[88] Focusing on a narrow subset of those emissions unsurprisingly means little revenue compared to a tax that covers all of Hungary's greenhouse gas emissions.

While the tax is designed with the EU ETS in mind, it targets a similar base. The free allocation of allowances is a hole in the EU ETS's base that the carbon tax addresses. However, the tax covers some emissions that were fully subject to the EU ETS.^[89] We chose to do the adjustment calculation regardless, but it only marginally improves Hungary's c-efficiency ratio.

Iceland

Iceland's Carbon Tax System

Rate (2023)	\$38.53 per ton
Total GHG Emissions (2023)	4.17 megatons of CO ₂ -equivalent
Actual Revenue (2023)	\$55.7 million
Potential Revenue	\$160.8 million
Unadjusted C-Efficiency Ratio	0.35 (12th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	1.81 megatons of CO ₂ -equivalent
Adjusted Potential Revenue	\$90.9 million
Adjusted C-Efficiency Ratio	0.61 (6th of 31)

Source: Author's calculations; World Bank Carbon Pricing Dashboard; EDGAR, "GHG Emissions of All Countries, 2024," September 2024; EU Emissions Trading System (ETS) Data Viewer.

Iceland introduced a carbon tax in 2010 in the aftermath of the Great Recession and global financial crisis, as the small nation faced a significant revenue shortfall.^[90] When the tax was introduced, the rate was set at around EUR 7 per ton of emissions, and it has slowly increased to around EUR 30 per ton in 2021.^[91] The country has further raised its carbon tax along with inflation.^[92]

Iceland's carbon tax has a comparatively high c-efficiency ratio. The primary driver of the high ratio is coverage of the transportation sector, with road transportation making up over 20 percent of overall emissions.^[93] Iceland taxes non-CO₂ emissions, such as F-gases, albeit at a lower rate (at least on a CO₂-equivalent basis). Most carbon taxes do not cover these emissions, so covering them all is a big boost. Slight variations in the CO₂ tax rate between different types of fossil fuel also reduce the tax's efficiency ratio.^[94]

Iceland's carbon tax is complementary with the EU ETS. The EU ETS covers aviation and shipping within the European Economic Area, and aviation and shipping are significant sources of emissions for the small island nation.^[95] As a result, subtracting the ETS's base leads to a dramatic improvement of the Icelandic carbon tax's c-efficiency ratio.

Ireland

Ireland's Carbon Tax System

Rate (2023)	\$52.74 per ton
Total GHG Emissions (2023)	57.85 megatons of CO2-equivalent
Actual Revenue (2023)	\$1.02 billion
Potential Revenue	\$3.05 billion
Unadjusted C-Efficiency Ratio	0.33 (13th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	12.19 megatons of CO2-equivalent
Adjusted Potential Revenue	\$2.41 billion
Adjusted C-Efficiency Ratio	0.42 (15th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer.

Ireland introduced a carbon tax in 2010 in the aftermath of the global financial crisis, as the banking-heavy economy experienced major tax revenue shortfalls.^[96] The tax is structured as several separate taxes on emissions: the Solid Fuel Carbon Tax, the Mineral Oil Tax, and the Natural Gas Carbon Tax.^[97] Ireland staggers its carbon tax increases. In October 2022, the carbon tax was raised to EUR 48.50 for auto fuels, but the tax remained at EUR 41.00 for all other fuels until May 2023. Similarly, the carbon tax went up to EUR 56 for auto fuels in October 2023.^[98]

Ireland’s carbon tax is focused on emissions from road transportation and commercial and residential buildings and exempts emissions from electricity production and industrial processes, both of which are largely covered by the EU ETS.^[99] This is a relatively common carbon tax base for European countries designing carbon taxes to complement the EU ETS, and subtracting the ETS emissions does improve the Irish carbon tax’s c-efficiency ratio.

However, the main factor reducing the Irish carbon tax’s c-efficiency ratio is not complementarity with the EU ETS. Instead, it’s the country’s composition of emissions: methane and nitrous oxide emissions from the agricultural sector constitute 38 percent of total GHG emissions, and neither gas is covered by the tax.^[100]

Japan

Japan's Carbon Tax System

Rate (2023)	\$2.17 per ton
Total GHG Emissions (2023)	1041.01 megatons of CO2-equivalent
Actual Revenue (2023)	\$1.67 billion
Potential Revenue	\$2.25 billion
Unadjusted C-Efficiency Ratio	0.74 (2nd of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.74 (2nd of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024.

Japan introduced a carbon tax in 2012. The tax rate was phased in, reaching 289 JPY per ton after three and a half years, and it is has remained at that level since.^[101] The tax is layered on top of several different energy-related excise taxes. In April 2023, Japan began a new set of carbon pricing initiatives, including a voluntary emissions trading system that will enter full-scale operation in 2027, along with a new, more substantial carbon tax.^[102]

Japan’s carbon tax has a high c-efficiency ratio due to its broad coverage of emissions across major sectors. Despite some exemptions for specific kinds of emissions within major sectors—e.g., fuel used in large-scale transportation such as passenger and cargo ships, railways, and aviation, and fuel used as inputs in industrial processes—the tax still covers a

substantial share of emissions in all major sectors.^[103] The Japanese carbon tax excludes non-CO2 greenhouse gasses like methane or nitrous oxide, but they make up less than 10 percent of Japan’s overall GHG emissions.^[104]

In addition to the national-level carbon tax, the city of Tokyo and Saitama Prefecture both employ cap-and-trade emissions schemes, covering 20 percent and 17 percent of localized greenhouse gas emissions, respectively.^[105] They do not come with exemptions for the national-level carbon tax, so there is no need for an adjustment calculation. Furthermore, while Japan is in the process of introducing an emissions trading scheme, it was still in the voluntary stage in 2023, so there is no need to adjust for that policy either.

Latvia

Latvia's Carbon Tax System

Rate (2023)	\$16.31 per ton
Total GHG Emissions (2023)	10.96 megatons of CO2-equivalent
Actual Revenue (2023)	\$8.2 million
Potential Revenue	\$178.7 million
Unadjusted C-Efficiency Ratio	0.05 (27th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	1.73 megatons of CO2-equivalent
Adjusted Potential Revenue	\$150.5 million
Adjusted C-Efficiency Ratio	0.05 (27th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer.

Introduced as part of the Natural Resources Tax Law in 2004, the Latvian carbon tax complements the EU ETS, focused on emissions in the power and industry sectors not covered by the EU-level policy.^[106] In 2017, the government increased the carbon and

energy tax rates to EUR 4.5 per ton of CO2.[107] The rate has been steadily increased, to EUR 9 per ton in 2020 and to EUR 15 per ton by the beginning of 2022, where it remained throughout 2023.[108]

Latvia’s low c-efficiency ratio is attributable to its narrow scope. It does not cover large portions of Latvian GHG emissions, like CO2 emissions from transportation (28 percent of 2023 emissions) and cumulative non-CO2 GHG emissions (40 percent of 2023 emissions). [109]

The carbon tax is merely designed to fill the gaps in the ETS base in the industrial and power sectors.[110] And even when adjusting for the emissions excluded thanks to the EU ETS, the Latvian tax does not perform well under the c-efficiency ratio, because most of Latvia’s GHG emissions do not come from industry and power generation.

Liechtenstein

Liechtenstein's Carbon Tax System

Rate (2023)	\$130.81 per ton
Total GHG Emissions (2023)	0.17 megatons of CO2-equivalent
Actual Revenue (2023)	\$4.96 million
Potential Revenue	\$21.58 million
Unadjusted C-Efficiency Ratio	0.23 (17th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.23 (19th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard.

The Liechtenstein carbon tax was implemented in 2008 in line with a bilateral treaty that requires the adoption of Swiss federal legislation on environmental taxes.^[111] Both countries levy the highest tax rate on carbon in Europe. The tax applies to carbon emissions from fossil fuel combustion in mainly the industry, power, buildings, and transport sectors. Liechtenstein also joined the EU ETS in 2021 and refunds carbon tax paid on fuel use already covered by the EU ETS or used merely in industrial processes. The resulting revenues are earmarked partially for environmental policy projects and partially for employer-side pension contributions.^[112] The government allows for full or partial tax refunds in exchange for emission reduction commitments or carbon offsets out to 2024.^[113]

Liechtenstein’s carbon tax has a broad scope but ends up with a below-average c-efficiency ratio. The country’s rank is reduced by full or partial tax refunds to selected businesses in exchange for commitments to emission reductions or using carbon offsets. Partial refunds to heat generation plants not covered by the EU ETS but meeting minimum environmental standards also reduce the c-efficiency ratio of Liechtenstein’s carbon tax.^[114]

We did not include an adjustment for the EU ETS’s application in Liechtenstein due to data availability issues.

Luxembourg

Luxembourg's Carbon Tax System

Rate (2023)	\$48.11 per ton
Total GHG Emissions (2023)	7.86 megatons of CO2-equivalent
Actual Revenue (2023)	\$294.9 million
Potential Revenue	\$378.2 million
Unadjusted C-Efficiency Ratio	0.78 (1st of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	0.89 megatons of CO2-equivalent
Adjusted Potential Revenue	\$335.4 million

Rate (2023)	\$48.11 per ton
Adjusted C-Efficiency Ratio	0.88 (1st of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer.

Luxembourg’s carbon tax (passed in 2019 and implemented in 2021) applies to CO2 emissions from fossil fuels used for transportation and heating. Emissions already covered by the EU ETS, as well as fuels used in industrial processes, are eligible for a tax refund.^[115] In addition to its carbon tax and the EU ETS, Luxembourg also levies separate fuel excise taxes.^[116] Imposition of the tax was accompanied by increased social support for low-income earners and some revenues are earmarked for a climate and energy fund. The basic rate gradually increased by 50 percent between 2021 and 2023. There is an increased rate for diesel which may be related to concerns over local pollution due to higher NO2 intensity.^[117]

Luxembourg has the highest carbon tax c-efficiency ratio in the world due to its broad coverage of emissions. The reluctance to use exemptions or selective incentives in its design mirrors the country’s approach to broad-based value-added taxes.^[118] The tax design is not perfect: the tax has slightly different rates for emissions from different fuels, and the tax does not cover non-CO2 emissions. However, that gap in the tax base does not substantially undermine the c-efficiency ratio because non-CO2 emissions account for only 10 percent of the country’s overall greenhouse gas emissions.^[119]

Luxembourg’s emissions profile also explains how the carbon tax can have such a high c-efficiency ratio while also being a complement to the EU ETS. Luxembourg’s carbon tax excludes emissions from the industrial and power sectors, but emissions from those sectors constitute a small share of the national total.^[120] Accordingly, subtracting out emissions covered by the ETS only slightly improves the country’s c-efficiency ratio.

Mexico

Mexico's Carbon Tax System

Rate (2023)	\$4.08 per ton
Total GHG Emissions (2023)	712.10 megatons of CO ₂ -equivalent
Actual Revenue (2023)	\$0.44 billion
Potential Revenue	\$2.90 billion
Unadjusted C-Efficiency Ratio	0.15 (21st of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.15 (21st of 31)

Source: Author's calculations; World Bank Carbon Pricing Dashboard; EDGAR, "GHG Emissions of All Countries, 2024," September 2024.

Mexico introduced a carbon tax in 2014. Additionally, Mexico introduced an ETS at the beginning of 2023.^[121]

Several reasons lead to Mexico's carbon tax's low c-efficiency ratio. For one, the tax is only on CO₂ emissions produced by fossil fuel activity in excess of emissions produced by natural gas, and natural gas emissions are entirely exempted.^[122] Given that natural gas is the source of more than half of Mexico's electricity generation, the exemption creates a significant hole in the tax base.^[123] The tax excludes non-CO₂ emissions like methane and nitrous oxide, which constituted 28 percent of the nation's emissions in 2023.^[124] Mexico further narrowed its carbon tax's base in March 2022, when the country suspended the tax on gasoline and diesel fuel (while also adding direct fiscal subsidies for fuel purchases).^[125]

Regarding tax interactions, Mexico has both a national-level ETS (which is still being rolled out) and a handful of subnational carbon taxes. The states of Mexico, Queretaro, Yucatan, Zacatecas, and Guanajuato all featured carbon taxes in 2023.^[126] However, a lack of data, combined with significant overlap in what emissions are covered by the national carbon tax, led us not to incorporate these other carbon prices into an adjusted c-efficiency ratio.

Northwest Territories

Northwest Territories' Carbon Tax System

Rate (2023)	\$48.03 per ton
Total GHG Emissions (2023)	1.36 megatons of CO2-equivalent
Actual Revenue (2023)	\$42.9 million
Potential Revenue	\$64.8 million
Unadjusted C-Efficiency Ratio	0.66 (4th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.66 (5th of 31)

Source: Author's calculations; World Bank Carbon Pricing Dashboard; Environment and Climate Change Canada Data Catalogue, "Environmental Provincial and Territorial Greenhouse Gas Emission Tables by IPCC Sector, 1990–2023."

The Canadian territory of Northwest Territories introduced a carbon tax in 2019 in response to the Canadian federal government’s carbon pricing backstop system, under which Canadian provinces or territories without sufficient carbon pricing mechanisms are subject to federal taxes or cap-and-trade systems.^[127] In 2023, the tax rate was CAD 65 per ton.

The Northwest Territories’ c-efficiency ratio is near the top of the rankings, although not the top as it was in the original edition of this report. The tax has a broad base, covering most CO2 emissions with limited direct exemptions, such as aviation fuel and fuel sold to native band governments.^[128] It also does not include non-CO2 emissions, but they account for under 10 percent of the province’s total emissions.^[129]

While Northwest Territories is a small jurisdiction population-wise, its carbon tax has several issues. A large portion of the revenue raised was used for direct offsets to the carbon tax burden, in a way that diluted the carbon price’s effect.^[130] The most salient example was the treatment of home heating oil. Initially, consumers of home heating oil received one-for-

one reimbursements for carbon taxes paid: functionally equivalent to an exemption but not reducing revenue on paper. Northwest Territories removed this rebate in April 2023, then reintroduced it in 2024.^[131]

In the context of Canada’s interlocking carbon pricing system, no price other than the Northwest Territories’ carbon tax applied in Northwest Territories in 2023, so there was no need to adjust the c-efficiency ratio. While Northwest Territories repealed the consumer side of its carbon tax in 2025, it maintained the portion of the tax focused on large industrial emitters, so the tax still exists in some form.^[132]

Norway

Norway's Carbon Tax System

Rate (2023)	\$94.49 per ton
Total GHG Emissions (2023)	56.72 megatons of CO2-equivalent
Actual Revenue (2023)	\$1.51 billion
Potential Revenue	\$5.36 billion
Unadjusted C-Efficiency Ratio	0.28 (16th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	21.33 megatons of CO2-equivalent
Adjusted Potential Revenue	\$3.34 billion
Adjusted C-Efficiency Ratio	0.45 (14th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer.

Norway introduced a carbon tax in 1991. Since the tax was introduced, the rate has been increased substantially, and exemptions have been slowly peeled back. In practice, the Norwegian carbon tax is broken up into several different components (all based on greenhouse gas emissions): a CO2 tax on mineral products (which includes mineral oil, petrol, gas, natural gas, and liquid petroleum gas), a CO2 tax on emissions from offshore

petroleum operations, a CO2 tax on emissions from waste incineration, and a tax on hydrofluorocarbons and perfluorocarbons (PFC)s.^[133] In local currency, the standard tax rate in 2023 was NOK 952 per ton.^[134]

Norway’s carbon tax has a roughly average c-efficiency ratio. The tax’s notable strengths are its coverage of transportation emissions (responsible for roughly a quarter of national GHG emissions) as well as a substantial chunk of non-CO2 emissions, such as methane and F-gases.^[135] Beyond those strengths, Norway’s carbon tax has many holes in its tax base thanks to its accommodations for the EU ETS.

Norway exempts many industrial emitters subject to the ETS from the carbon tax. The exception is the oil and gas extraction industry. Emissions from offshore oil and gas extraction (also representing a quarter of Norway’s total) are subject to both the ETS and a reduced carbon tax rate.^[136] The combined ETS and reduced carbon tax rate can mean Norwegian offshore oil drilling emissions are subject to among the highest carbon prices in the world.

The Norwegian approach to managing the domestic carbon tax and the EU ETS complicates our methodology. The tax exempts a large portion of ETS emissions and provides a reduced rate when the bases overlap. Accordingly, the unadjusted c-efficiency ratio is unfairly harsh, while the adjusted c-efficiency ratio is too generous.

Poland

Poland's Carbon Tax System

Rate (2023)	\$15.32 per ton
Total GHG Emissions (2023)	363.79 megatons of CO2-equivalent
Actual Revenue (2023)	\$0.01 billion
Potential Revenue	\$5.57 billion
Unadjusted C-Efficiency Ratio	0.00 (31st of 31)

Rate (2023)	\$15.32 per ton
GHG Emissions Subject to Other, Complementary Carbon Price	152.45 megatons of CO ₂ -equivalent
Adjusted Potential Revenue	\$3.24 billion
Adjusted C-Efficiency Ratio	0.00 (31st of 31)

Source: Author's calculations; World Bank Carbon Pricing Dashboard; EDGAR, "GHG Emissions of All Countries, 2024," September 2024; EU Emissions Trading System (ETS) Data Viewer.

Poland's carbon tax was one of the world's earliest, although it receives much less attention than the other carbon taxes introduced in the early 1990s. Poland's carbon tax is often forgotten in the discussion of early carbon taxes because its tax is typically called a national emissions fee, not a carbon tax.^[137] Poland has a series of fees that cover several forms of environmental pollution, fees that in some cases predate democratization.^[138] While not billed as carbon taxes, the emissions-related fees do qualify, as the fees are levied according to emission volume, rather than on the value of goods sold. The emissions fees are just a few of the country's long list of pollution and resource fees.^[139]

Poland's rock-bottom c-efficiency ratio reflects the unusual structure of its emissions pricing. The tax rate on carbon dioxide is roughly \$0.08 per ton, but the tax rate on F-gases is \$15.32 per ton. The taxes on other greenhouse gas emissions, such as methane and nitrous oxide, do not reach above \$1 per ton.^[140] Using the highest rate on greenhouse gas emissions means using the F-gas tax rate, rather than the CO₂ tax rate. F-gases constitute under 2 percent of overall Polish GHG emissions, so the c-efficiency ratio is extremely low because even if other GHG emissions are technically covered, they are practically exempt relative to F-gases.^[141]

These handful of fees are intended as complements to the EU ETS. The EU ETS does cover a large portion of Polish GHG emissions, but subtracting out those emissions does not substantially improve Poland's c-efficiency ratio. Polish GHG emissions pricing would benefit from a more uniform approach.

Portugal

Portugal's Carbon Tax System

Rate (2023)	\$28.73 per ton
Total GHG Emissions (2023)	53.00 megatons of CO2-equivalent
Actual Revenue (2023)	\$0.49 billion
Potential Revenue	\$1.52 billion
Unadjusted C-Efficiency Ratio	0.32 (14th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	12.76 megatons of CO2-equivalent
Adjusted Potential Revenue	\$1.16 billion
Adjusted C-Efficiency Ratio	0.42 (16th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer.

Portugal implemented a carbon tax in 2015, as an addition to its existing petroleum and energy products tax. Portugal’s carbon tax rate is based on average ETS prices over the preceding two years as well as a target price of EUR 25 per ton.^[142] Portugal also added a flat EUR 2 tax on international air and sea travelers on international aviation and shipping. While this particular tax is referred to as a carbon tax, it is not directly based on the emissions content of travel activity.^[143]

Portugal’s carbon tax has a roughly middle-of-the-road c-efficiency ratio. The tax primarily applies to the building and transportation sectors, with the latter producing roughly 30 percent of the country’s GHG emissions.^[144] On the other hand, the tax does not cover non-CO2 GHGs like methane and nitrous oxide, which also combine for roughly 30 percent of the country’s GHG emissions.^[145] There are some slight variations in tax rates for different taxable activities, with the highest (in theory) falling on coal, and coal use has all but ceased in Portugal as of 2023.^[146]

The Portuguese carbon tax is in practice complementary with the EU ETS. Coal-powered electricity generators paid both the carbon tax and the EU ETS when those generators were active, but today overlap between the two prices is negligible.^[147] Excluding emissions subject to the ETS improves Portugal’s c-efficiency ratio.

Singapore

Singapore's Carbon Tax System

Rate (2023)	\$3.77 per ton
Total GHG Emissions (2023)	74.29 megatons of CO2-equivalent
Actual Revenue (2023)	\$162.2 million
Potential Revenue	\$279.9 million
Unadjusted C-Efficiency Ratio	0.58 (5th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.58 (8th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024.”

Singapore introduced a carbon tax in 2019 at a rate of SGD 5 per ton. The rate was still SGD 5 in 2023, but the city-state raised the rate from SGD 5 to SGD 25 per ton in 2024 and plans to further increase the rate to SGD 45 in 2026 and eventually to SGD 80 per ton by 2030. ^[148] The tax has a high de minimis threshold of 25,000 metric tons of CO2 emissions per year and a small number of taxpayers, approximately 50, exceed the threshold. ^[149] The tax covers industrial and power emissions (including many non-CO2 GHG emissions), while excluding transportation emissions.^[150]

Singapore’s high c-efficiency ratio, even with a relatively high de minimis exemption threshold, shows how an upstream carbon tax can cover a large share of emissions in a jurisdiction. Including non-CO2 emissions in the tax base is a particularly important policy

choice for Singapore, as they made up 23 percent of Singapore’s emissions in 2023. Conversely, excluding the transportation sector does not have a substantially negative impact on the tax’s efficiency ratio, as transportation emissions are less than 10 percent of the nation’s emissions.^[151]

In 2024 and beyond, under the increased carbon tax rate, the elevated carbon tax includes some carbon offset credits (limited to just 5 percent of carbon tax liability) as well as potential transitional rebates for oil refiners and pharmaceutical manufacturers.^[152] These new policies will likely reduce the tax’s c-efficiency ratio.

South Africa

South Africa's Carbon Tax System

Rate (2023)	\$8.93 per ton
Total GHG Emissions (2023)	522.12 megatons of CO2-equivalent
Actual Revenue (2023)	\$0.13 billion
Potential Revenue	\$4.66 billion
Unadjusted C-Efficiency Ratio	0.03 (29th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.03 (30th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024.”

In June 2019, South Africa became the first African country to introduce a carbon tax. The carbon tax is aimed at large businesses across economic sectors, including industry, power, and transportation.^[153] The first phase of the tax (currently in law) is transitional, with substantial consideration for reducing the impact on low-income households and trade-

exposed industries. Originally, the transitional phase was supposed to last until the end of 2022, but the 2022 budget extended the first phase of the policy until the end of 2025.^[154] In 2023, the primary tax rate was ZAR 159 per ton.^[155]

On paper, South Africa’s carbon tax seems well-designed, with around 80 percent of emissions included in the tax base.^[156] That fact makes the country’s low c-efficiency ratio puzzling. However, the South African carbon tax’s system of tax-free allowances explains its low performance. Under the allowances, between 60 and 75 percent of emissions in certain sectors end up effectively untaxed, with some additional exemptions meaning up to 95 percent of a sector’s emissions can go untaxed.^[157] Once the transitional period is over, and certain exemptions phase out, the carbon tax’s c-efficiency ratio should improve.

Spain

Spain's Carbon Tax System

Rate (2023)	\$16.31 per ton
Total GHG Emissions (2023)	285.38 megatons of CO2-equivalent
Actual Revenue (2023)	\$0.10 billion
Potential Revenue	\$4.66 billion
Unadjusted C-Efficiency Ratio	0.02 (30th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	81.12 megatons of CO2-equivalent
Adjusted Potential Revenue	\$3.33 billion
Adjusted C-Efficiency Ratio	0.03 (29th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer.

Spain introduced a tax on fluorinated gas emissions in 2014.^[158] Enforcement challenges led to significant administrative reforms in 2022.^[159] The Spanish government revised the tax’s administration, making manufacturers, importers, and other intermediate firms, as

opposed to final retail sellers, responsible for remitting the tax.^[160] The reform eliminates some small exemptions, while also introducing some small new ones. In 2023, the rate was just under EUR 15.

The Spanish tax performs poorly under the c-efficiency ratio because it does not tax the major greenhouse gases. CO2 made up 76 percent of GHGs in Spain in 2023, with methane making up another 16 percent; f-gases are a small fraction of total GHG emissions, and Spain is no exception.^[161] The EU ETS covers Spanish CO2 emissions from power generation and some industry, but CO2 emissions from the transportation sector (the largest sector in terms of greenhouse gas emissions in Spain) are not covered.^[162] Accordingly, even adjusting for the ETS’s coverage does not significantly improve the Spanish tax’s c-efficiency ratio.

Sweden

Sweden's Carbon Tax System

Rate (2023)	\$142.94 per ton
Total GHG Emissions (2023)	49.12 megatons of CO2-equivalent
Actual Revenue (2023)	\$2.17 billion
Potential Revenue	\$7.02 billion
Unadjusted C-Efficiency Ratio	0.31 (15th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	17.01 megatons of CO2-equivalent
Adjusted Potential Revenue	\$4.59 billion
Adjusted C-Efficiency Ratio	0.47 (13th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer.

Sweden’s carbon tax is one of the oldest in the world. Introduced in 1991, the carbon tax has steadily increased from SEK 250 per ton of CO2 in 1991 to SEK 1,300 in 2023, with the largest increases coming in the early 2000s.^[163] The policy was originally introduced in a broader tax reform package that reduced Sweden’s relatively high corporate and personal income tax rates.^[164] In addition to a carbon tax, Sweden has several targeted environmental and energy-related excise taxes and participates in the EU ETS.^[165]

Sweden’s middle-of-the-pack c-efficiency ratio can be explained by scope. On the positive side, the tax covers the transportation sector (responsible for 27 percent of the nation’s GHG emissions) with limited exceptions.^[166] The tax has reduced emissions in the sector.^[167] On the other hand, the tax does not cover non-CO2 GHG emissions like methane and nitrous oxide, which make up 28 percent of Sweden’s emissions as of 2023.^[168]

The tax is complementary with the EU ETS, and it accordingly excludes the 17.01 megatons (roughly 35 percent of Swedish emissions) the cap-and-trade system covers.^[169] After taking the ETS into account, the Swedish c-efficiency ratio improves substantially, both in absolute terms and in the ranking.

Switzerland

Switzerland's Carbon Tax System

Rate (2023)	\$130.81 per ton
Total GHG Emissions (2023)	43.45 megatons of CO2-equivalent
Actual Revenue (2023)	\$1.17 billion
Potential Revenue	\$5.68 billion
Unadjusted C-Efficiency Ratio	0.21 (19th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	5.5 megatons of CO2-equivalent
Adjusted Potential Revenue	\$4.96 billion
Adjusted C-Efficiency Ratio	0.23 (18th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024; EU Emissions Trading System (ETS) Data Viewer; International Carbon Action Partnership, “Switzerland Emissions Trading System,” accessed December 2024, <https://icapcarbonaction.com/en/ets/switzerland-emissions-trading-system>.

Switzerland levies the highest tax rate on carbon emissions in Europe, at EUR 120.16 per ton (CHF 120 per ton) as of 2023.^[170] The Swiss carbon tax was introduced in 2008 at a tenth of the current rate. The tax applies to CO2 emissions from fossil fuels used to generate heat, light, or electricity. In addition to its carbon tax, Switzerland operates its own ETS, which has been closely linked to the EU ETS since 2020 and covers the same sectors.^[171] Emissions covered by the Swiss ETS are exempt from the Swiss carbon tax.

The Swiss carbon tax ranks in the middle among the surveyed jurisdictions. In terms of strengths, the Swiss carbon tax covers emissions from heating, which represent a sizeable share of the country’s GHG emissions. On the other hand, it does not include the transportation sector. Instead, the transportation sector is subject to the Mineral Oil Tax, an excise tax on different forms of gasoline.^[172] Switzerland also utilizes some other exemptions for specific energy-intensive industries.^[173] The tax is also based on CO2, so it excludes methane, nitrous oxide, and other greenhouse gases, and those pollutants make up roughly 20 percent of the country’s total GHG emissions as of 2023.^[174]

The tax is complementary with Switzerland’s ETS. However, Switzerland’s ETS has a relatively narrow base compared to many other ETS systems, covering a small share of Swiss emissions, so subtracting out the emissions subject to it does not provide a particularly large boost to the country’s c-efficiency ratio.^[175]

Ukraine

Ukraine's Carbon Tax System (2021)

Rate (2023)	\$0.82 per ton
Total GHG Emissions (2023)	216.09 megatons of CO2-equivalent
Actual Revenue (2023)	\$85.9 million
Potential Revenue	\$177.3 million

Rate (2023)	\$0.82 per ton
Unadjusted C-Efficiency Ratio	0.48 (7th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.48 (12th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024.

Ukraine first introduced a carbon tax in 2011. Throughout most of the carbon tax’s existence, the rate has been low—well below USD 1 per ton of CO2.^[176] In early 2022, Ukraine raised its carbon tax from UAH 10 to UAH 30 per ton (approximately 1 USD), where it has remained. Ukraine is considering implementing an ETS as well.

While Ukraine imposes a very low carbon tax rate, the broad carbon tax base covering CO2 emissions, such as emissions from industrial activities, electricity and heat generation, and residential and commercial buildings, leads it to perform well under the c-efficiency ratio.^[177] However, the tax excludes CO2 emissions from transportation, as well as non-CO2 GHG emissions, the latter of which make up over a third of the nation’s total emissions as of 2023.^[178]

Another factor pushing down the c-efficiency ratio is the Russian invasion of Ukraine. Ukraine’s total greenhouse gas emissions include emissions from occupied territories, emissions which the Ukrainian government cannot effectively tax.^[179] The tax base is fairly broad, but not all-encompassing. Given those two factors, the relatively high c-efficiency ratio is a testament to effective tax collection in Ukrainian territory. That said, under these circumstances, the estimated national GHG emissions come with significant uncertainty.

United Kingdom

United Kingdom's Carbon Tax System

Rate (2023)	\$22.28 per ton
Total GHG Emissions (2023)	379.32 megatons of CO ₂ -equivalent
Actual Revenue (2023)	\$0.99 billion
Potential Revenue	\$8.45 billion
Unadjusted C-Efficiency Ratio	0.12 (22nd of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.12 (22nd of 31)

Source: Author's calculations; World Bank Carbon Pricing Dashboard; EDGAR, "GHG Emissions of All Countries, 2024," September 2024.

The United Kingdom introduced a carbon tax (known as the Carbon Price Support) in 2013. As the EU ETS's price had been low and volatile, the British government introduced the carbon tax to provide a stable top-up incentive for reducing carbon emissions.^[180] Since Brexit, the UK has established its own ETS system, and the Carbon Price Support provides an additional incentive on top of that new ETS. The Carbon Price Support has remained at GBP 18 for the past several years.^[181]

The tax is solely focused on the power sector, ignoring larger sectors of UK greenhouse gas emissions like transportation.^[182] Since the carbon tax was introduced, CO₂ emissions from the power sector have shrunk dramatically in both absolute terms (from 168.2 megatons in 2013 to 53.9 megatons in 2023) and as a share of the UK's overall GHGs (from 30 percent in 2013 to 14 percent in 2023).^[183] This trend was in part driven by the carbon tax pushing the retirement of coal power plants.^[184] In this respect, the United Kingdom's low c-efficiency score partly reflects its successes as a policy.

Unlike many other carbon taxes in Europe, the UK carbon tax is not designed as a complement to an ETS. Instead, the UK carbon tax is layered on top of largely the same emissions the UK ETS covers. The UK ETS has a slightly broader scope than the carbon tax,

as it covers some emissions from industrial activities and aviation, but the bases of the two policies are decidedly not complementary in that they stack on top of each other.^[185] As a result, it does not make sense to adjust the tax’s c-efficiency ratio.

Uruguay

Uruguay's Carbon Tax System

Rate (2023)	\$155.87 per ton
Total GHG Emissions (2023)	41.63 megatons of CO2-equivalent
Actual Revenue (2023)	\$0.27 billion
Potential Revenue	\$6.49 billion
Unadjusted C-Efficiency Ratio	0.04 (28th of 31)
GHG Emissions Subject to Other, Complementary Carbon Price	N/A
Adjusted Potential Revenue	N/A
Adjusted C-Efficiency Ratio	0.04 (28th of 31)

Source: Author’s calculations; World Bank Carbon Pricing Dashboard; EDGAR, “GHG Emissions of All Countries, 2024,” September 2024.

Uruguay passed a carbon tax in November 2021, which took effect on January 1, 2022, through Presidential Decree No. 441/021.^[186] The tax applies to all liquid fuels, with the exemption of jet fuel and sales of fuel to gasoline manufacturers. Decree 435/022 set the new value of the Uruguayan carbon tax at UYU 6,024 per ton for 2023, and the tax was raised further in 2024.^[187]

The biggest hole in Uruguay’s carbon tax base is not including non-CO2 emissions. Almost 80 percent of Uruguay’s emissions are non-CO2 gases like methane and nitrous oxide. Methane and nitrous oxide emissions from agriculture alone account for two-thirds of Uruguay’s total emissions.^[188] This provides the primary explanation for the country’s carbon tax performing poorly on the c-efficiency ratio. The focus on gasoline means the tax

covers transportation emissions well, and transportation produces the most CO₂ emissions of any sector in Uruguay, but it misses CO₂ emissions produced in other sectors, such as industrial processes.[189]

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[22] Shuting Pomerleau, "Administrative Costs of a Carbon Tax."

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[24] Ibid., European Commission, "Global Greenhouse Gas Emissions," EDGAR v7.0, 2022, https://edgar.jrc.ec.europa.eu/dataset_ghg70.

[25] The c-efficiency ratio should not be sensitive to exchange rate fluctuations, as revenues and rates are both adjusted by the same exchange rate.

[26] World Bank, "Carbon Pricing Dashboard."

[27] Consider a country that had a carbon tax of \$10 per ton from January 1st to June 30th that raised the tax rate to \$20 in July. If they produced 100 tons of emissions in that year (and the tax was perfectly administrated and all-encompassing) they would raise \$1,500 in

carbon tax revenue. Using the tax rate as of April 1, this country would have an impossible carbon efficiency ratio of 1.5, as their potential revenue calculated using the April 1 rate would be \$1,000.

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